Customising Gymnasium Environments and Implementing Reinforcement Learning Agents with Stable-Baselines3

Overview

In this assignment, you will develop a Reinforcement Learning agent using an existing Gymnasium (https://gymnasium.farama.org/) environment as a basis. **Your task is to introduce specific changes or customisations to the environment and train a reinforcement learning agent using the Stable Baselines library. The goal is to evaluate how these changes affect the learning process and the agent's performance.**

Milestones

Week 1: Customising the Environment

- Select an Existing Gymnasium Environment: Choose an existing environment that interests you. It could be a classic control task, a robotic simulation, or a custom environment.
- Identify Necessary Modifications: Carefully analyse the chosen environment and identify areas where you can make significant modifications or customisations. **These changes may include altering rewards, introducing additional challenges, or modifying the state space.**
- Implement the Customisations: Implement the identified customisations in the environment code. Ensure that the modified environment is compatible with the Gymnasium interface and maintains a reasonable level of complexity for reinforcement learning experimentation.

Week 2: Reinforcement Learning Agent

- Select a Reinforcement Learning Algorithm: Choose a reinforcement learning algorithm from the Stable-Baselines3 library. Popular options include Proximal Policy Optimization (PPO), Deep Deterministic Policy Gradients (DDPG), or any other algorithm you consider suitable for your customised environment.
- Implement the Agent: Write code to create and train a reinforcement learning agent using your chosen algorithm. Be sure to integrate the agent with your customised environment.
- Hyperparameter Tuning: Experiment with different hyperparameters to fine-tune your agent's performance. Document the changes you make and the reasons behind them.

Week 3: Evaluation and Analysis

- Training and Evaluation: Train your reinforcement learning agent in the customised environment and evaluate its performance using appropriate metrics. Be sure to run multiple experiments to obtain statistically significant results.
- Compare Results: Compare the performance of your customised environment with the original, unmodified Gymnasium environment. Analyse how your changes affected the agent's learning process and final performance.
- Prepare a Presentation: Summarise your findings. Include details about the environment modifications, the reinforcement learning algorithm used, hyperparameter choices, and the comparative results. Discuss the implications of your customisations on the agent's performance.

Last modified: Wednesday, 7 January 2026, 23:43

◀ Assignment 1: Presentation Slots

Assignment 2: Submissions ▶